

FVN Research

Audit Controls Report

Controls, gates, lineage, contracts, release package, and freeze evidence.

Audit architecture

- End-to-end smoke runner: verifies the critical chain on deterministic fixture data.
- Live-data readiness checker: blocks live execution when compliance or source inputs are missing.
- Live pipeline wrapper: logs every command with readiness evidence.
- Data lineage graph: machine-readable nodes and edges from source to reports.
- Schema contract registry: required columns and quality gates for critical artifacts.
- Release checklist: one pre-publication gate combining verdict, smoke, schema, lineage, reproduction, and live readiness.
- Publication package: sanitized public release excluding raw/private/intermediate data.
- Final archive freeze: immutable metadata and checksums for library storage.

Validation status

The final archive freeze implementation was validated with config check, skeleton audit, and the full available test suite: 192 passed.

Known warnings

- Ablation test fixtures can trigger constant-input correlation warnings; this is expected for synthetic constant predictions.
- A pandas FutureWarning appears in one walk-forward split test fixture for month-end frequency notation.
- These warnings did not fail the suite and do not indicate a broken pipeline.

Reviewer conclusion

The archive is suitable for FVN Library storage as a validated research infrastructure and audit package. A live empirical publication should be created only after source data readiness is complete and the live pipeline has generated fresh evidence.